

A drone-based vision system to identify crop diseases by analyzing leaf patterns and discolorations.

Tarun Kumar
tarun.12215310@lpu.in

*Lovely Professional University School of
Computer Science and Engineering*

Rishav Vashisht
rishav.vashisht@lpu.in
*Lovely Professional University
School of Computer Science and Engineering*

Abstract—Food safety and plant health are interlinked. All people are concerned with the health of green plants. Plant diseases interrupt or alter an important activity of a plant by disturbing its usual state. The suggested method helps in identifying plant diseases.

The database collected from the Internet is properly partitioned, and the different plant species are identified to get a test database having many plant diseases that are utilized to check the correctness and confidence of the project. Then, based on training data, we will train our classifier, and the output will be predicted with maximum possible accuracy. We utilize a Convolution Neural Network (CNN), which consists of multiple layers utilized in the prediction. CNN's outperform other technologies in crop disease categorization and detection problems. They are capable of addressing challenging conditions with difficult imaging. A sample drone model is being used to monitor vast farming fields in real time, where a high-definition camera is mounted to capture pictures of the plants, which would be used as input to ascertain if the plant is healthy or not.

Index Terms—Convolution Neural Networks, Diseases, Feature Extraction, Agriculture, crops, Image Classification, object detection.

I. INTRODUCTION

Agriculture is a crucial sector in India, as it is the primary source of living for most of the population. It is one of the major contributors to the economy of the country, providing around 16% of India's GDP and employing nearly 50% of the workforce of the country. Agriculture is also a key player in fulfilling the food and nutritional requirements of the increasing population. Also, agricultural product export is a significant foreign exchange earner for India. In short, the significance of agriculture in India cannot be exaggerated. Currently, leaf diseases are a serious issue that have a detrimental effect on plant longevity and the yield of high-quality crops. Also, relying solely on your eyes to decide the current state of the leaf is very difficult. This reduces the yield of the crops of better quality. We wanted to employ a deep learning-based approach for segmentation, to select each tiny portion of the leaf, and to identify the disease, along with assessing the quality of the plant, so that this problem can be rectified. Data collection, processing, and input application rate variability are the core operations. Plant health and food safety are closely interconnected. Plant health is a widely used but poorly defined concept. Image processing can be considered as a subarea of signal

processing. The input to image processing is in the form of an image, i.e., photograph or video. Image processing will yield either an image or a collection of features or measures

appropriate to the input image. Naked-eye inspection is the conventional method for detecting and recognizing plant diseases, which is a slow and imprecise process. Because of the lack of experience in certain nations, reaching experts to diagnose plant disease takes time and money. Disciplined inspection of plants leads to the formation of numerous diseases on the plant, which requires the application of more chemicals for healing; moreover, these chemicals are toxic to other animals, insects, and birds that are agricultural-friendly. Automatic plant disease detection is essential for identifying symptoms of diseases at their initial stages, when they appear on a plant's developing leaf. Deep learning algorithms have been trending in recent years for image classification problems. Due to advances in artificial intelligence research, it is now feasible to diagnose plant diseases automatically from raw images. A learning approach based on neural networks is referred to as deep learning[2]. This training has the benefit of being capable of automatically extracting features from images. The neural network is trained to learn feature extraction. So, by minimizing the biotic causes of high agricultural yield losses, we can enhance the productivity and quality of crops. There are various machine learning and deep learning techniques that can be employed for this.

II. RELATED WORKS

Here in this section the existing research works concerning image segmentation and classification in the area of crop disease detection are described. Genetic algorithm is utilized to segment colored images in this method. The method is a way of representing an image as a weighted undirected graph where edges indicate similar pixels and vertices represent pixels. Brightness, color, and texture richness are taken into account while measuring the similarity between two pixels. Application of good procedures to determine healthy and diseased leaves helps in controlling crop loss and productivity. This chapter is a compilation of existing machine-learning methods for plant disease identification.

A. Shape and Texture-Based Identification

The authors of [1] utilized tomato-leaf images to detect diseases. They categorized ill segments based on various geometric and histogram-based features and an SVM classifier with different kernels. S. Kaur et al. [2] found three different soybean diseases based on colour and textural features.

In [3] P Babu et al. detected plant leaves and diseases using a

feed-forward neural network and back-propagation. S. S. Chouhan et al. [4] detected plant leaves and fungal infections with a bacterial-foraging-optimization-based radial-basis function neural network (BRBFNN). They utilized a region-growing algorithm to extract information from a leaf based on seed points with similar qualities in their methods. The bacterial-foraging optimization method is utilized to enhance classification accuracy and accelerate a network.

B. Segmentation using Traditional Methods

In paper [5], Vijai Singh presented a photo segmentation system that is employed for automatic diagnosis and classification of plant leaf diseases. The genetic method employed for picture segmentation, an important step in plant leaf disease identification. The average classification accuracy of the proposed algorithm, which was employed to finish the classification, is 97.6%. The authors of [6] suggested a three-step segmentation process. On green leaves, the non-differentiated disease patches should be cut off first. The fragmented image is then converted into 8-bit grayscale images using single thresholding after some odd characteristics are detected using a grey histogram. Lastly, compare the size and stems of the ill spots, and afterwards use area thresholding to separate the corresponding images.

C. Deep-Learning-Based Identification

Mohanty et al. detected 26 different plant diseases with AlexNet and GoogleNet CNN architectures. Ferentinos et al. [7] recognized 58 unique plant diseases employing various CNN architectures with a high degree of classification accuracy. They also experimented with the CNN architecture with actual photographs in their approach. Sladojevic et al. [8] designed a deep learning system for the identification of 13 unique plant diseases. They trained the CNN using the Caffe DL framework. The authors of [9] proposed a nine-layer CNN model to identify plant diseases. They used the PlantVillage dataset and data-augmentation methods to increase the data quantity for experimentation purposes and subsequently analyzed performance.

D. Leaf Image Classification

Based on a paper [8] study, segmenting and identifying diseases in live images are required for cardamom plant leaf disease detection. The proposed methodology's U2-Net architecture produces results without compromising the original image quality by eliminating the detailed background. EfficientNetV2-S and EfficientNetV2-L attained detection accuracy for the cardamom plant dataset of 98.28% and 98.26%, respectively. In work [10], scientists introduced NAS-Unet, which is stacked by the equal number of DownSC and UpSC on a U-like skeleton network. For speeding up the search, insert a U-shaped skeleton and the memory-efficient

Binary gate search approach. DownSC and UpSC are cell designs for semantic picture segmentation in this case.

III. PROPOSED METHODOLOGY

There are several various ways of detecting plant diseases through machine learning, but most of them consist of applying a mixture of image processing and machine learning algorithms to determine the existence of diseases in

plants. One typical method is applying convolutional neural networks (CNNs) to process images of plants and determine the presence of diseases. The CNNs are normally trained on a huge set of labeled images where the labels denote whether a specific plant in the image is healthy or infected.

A. Dataset

Within the PlantVillage dataset there are 38 plant crop-disease combinations, ranging across 14 crop species and 26 diseases, present in the 54,305 leaf images as presented in Fig 1. The high-quality photos of plant leaves in this dataset each contain one leaf. Also, it has pre-determined training and testing subsets, which we utilized within our research.

Classification Report:				
	precision	recall	f1-score	support
Pepper_bell_Bacterial_spot	0.75	0.92	0.82	199
Pepper_bell_healthy	0.96	0.85	0.90	295
Potato_Early_blight	0.97	0.85	0.91	200
Potato_Late_blight	0.77	0.88	0.82	200
Potato_healthy	0.95	0.63	0.76	30
Tomato_Bacterial_spot	0.96	0.95	0.96	425
Tomato_Early_blight	0.76	0.69	0.72	200
Tomato_Late_blight	0.89	0.79	0.84	381
Tomato_Leaf_Mold	0.84	0.84	0.84	190
Tomato_Septoria_leaf_spot	0.82	0.88	0.85	354
Tomato_Spider_mites_Two_spotted_spider_mite	0.85	0.90	0.87	335
Tomato_Target_Spot	0.85	0.81	0.83	280
Tomato_Tomato_YellowLeaf_Curl_Virus	0.97	0.95	0.96	641
Tomato_Tomato_mosaic_virus	0.88	0.95	0.91	74
Tomato_healthy	0.91	0.99	0.95	318
accuracy			0.88	4122
macro avg	0.87	0.86	0.86	4122
weighted avg	0.89	0.88	0.88	4122

Fig. 1. Overview of PlantVillage Dataset

It was created through six different augmentation strategies to generate more diverse datasets with different environmental conditions. Scaling, rotation, noise injection, gamma correction, picture flipping, and PCA colour augmentation were some of the augmentations utilized in this process.

B. Image Processing

The dataset must be resized owing to the potential of an imbalance in size in images as a result of the size of the dataset. To smooth noise and enhance the image, gaussian blur is applied subsequent to downsizing the raw input photos to 256 x 256. Gaussian blur is a blurring effect widely applied in image processing and computer vision. Since images are two dimensional, in 2-D form, a Gaussian kernel is written as 1. The Gaussian blur filter operates by convolving the original image with a Gaussian function. The function is a bell-shaped curve centered at the origin, meaning that the filter applies a uniform blurring effect to the whole image.

$$G2D(a, b, \sigma) = \frac{1}{2\pi\sigma^2} e^{-\frac{a^2+b^2}{2\sigma^2}}$$

where a and b are the indices of location, σ is the standard deviation of the distribution, and The variance of the Gaussian distribution, which determines the degree of the blurring effect around a pixel, is regulated by σ 's value.

The degree of blurring can be regulated by varying the standard deviation of the Gaussian function, which controls the spread of the bell-shaped curve. Generally, Gaussian blur is a handy technique for image smoothing and noise reduction. It is commonly applied in computer vision algorithms to pre-process images prior to applying more complicated operations. It is also widely utilized in computational photography, where it can be employed to produce effects such as depth of field and motion blur.

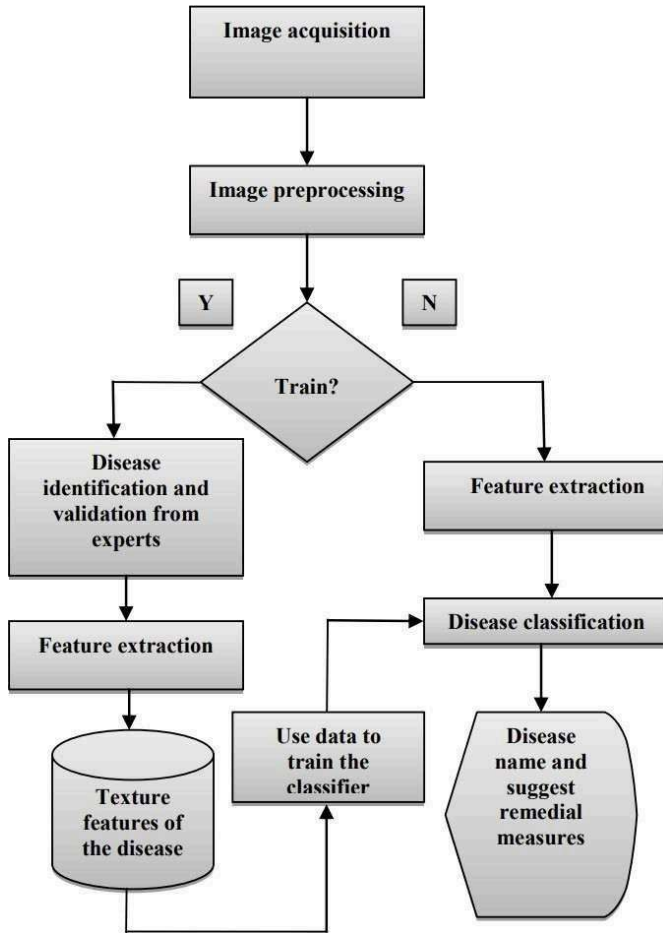


Fig. 2. Block diagram of proposed work

By employing different processing methods or combinations of several processing, i.e., random rotation, shifts, shear, and flip, etc., image augmentation artificially creates training images. The ImageDataGenerator API in Keras makes it

easy to create an improved picture generator. A tool for real time data augmentation named ImageDataGenerator generates batches of pictures data.

DATA AUGMENTATION TECHNIQUES	
Technique	Hyperparameter
Scaling	Yes
Rotation	Yes
Noise injection	Factor = 1
Random Flip	Vertical
Gamma correction	Yes
PCA colour	3 x 3

Steps involved in preparing input images for the classification of healthy and sick leaves using a neural network include:

- Grayscale conversion of images is often performed to emphasize the form and texture of the leaves instead of their color and to minimize the number of input dimensions. pixel values in an image are normalized.
- Resize the images into a uniform scale (254 x 254) suitable for the CNN model. This is often done to limit the computational demand of training the model and to have all images of the same size, which is critical to ensure that the CNN can effectively analyze the images.
- Image pixel values are normalized. This is often performed in order to boost the performance of the CNN by making sure the pixel values lie within a regular range (e.g., 0 and 1) and have zero mean and unit variance.
- Image augmentations are added to the images. This involves adding random transformations to the images (e.g., rotation, scaling, etc.) in an attempt to create more training data and make the model more resilient.

The block diagram in Figure 2 provides an overview of the suggested methodology.

1) *Image Thresholding*: With image thresholding, an image can be binarized based on pixel intensities. This type of thresholding process usually accepts as input a grayscale image and a threshold. What is produced are binary images. If the input pixel's intensity is greater than a threshold, the corresponding output pixel is marked as white (foreground), and if it is less than or equal to the threshold, it is marked as black (background). To calculate the spread for the pixel levels on both sides of the threshold, or the pixels in the foreground or background, Otsu's thresholding method loops through all possible threshold values. The goal is to find the threshold value at which the foreground and background spreads sum up to the minimum.

$$\sigma_w^2(t) = \omega_1(t)\sigma_1^2(t) + \omega_2(t)\sigma_2^2(t) \quad (2)$$

The entire computation formula of Otsu's thresholding can be expressed by Eq 2, where weighted variance of classes represented by $\sigma^2(t)$. $\omega_1(t)$ and $\omega_2(t)$ are the probabilities of the two classes divided by a threshold t .

2) *Canny Edge Detector*: Edge-based segmentation is an image processing technique for detecting and extracting the edges of objects within an image. This is usually achieved by applying a filter to the image that highlights the edges and boundaries between various regions of the image, like the edges between foreground and background objects. The result of an edge-based segmentation algorithm is a collection of edge maps, which are binary images that indicate the positions of edges in the input image. These edge maps may then be fed as input to other image processing algorithms, including object recognition and tracking algorithms, to detect and classify the objects in the image. A multi-stage edge detector is the Canny filter. A Gaussian derivative filter is applied to find the intensities of the gradients. The Gaussian smooths the quantity of noise within the image. Edge pixels are retained or eliminated by hysteresis thresholding applied on the gradient magnitude. The Canny has three parameters that can be changed: the width of the Gaussian (the wider the Gaussian, the noisier the image), and the low and high threshold for hysteresis thresholding. The input images can then be passed to the CNN model for training and testing after pre-processing.

C. Training

The preprocessed are now splitted into train test splits in the ratio of 80:20 and after splitting the training is being

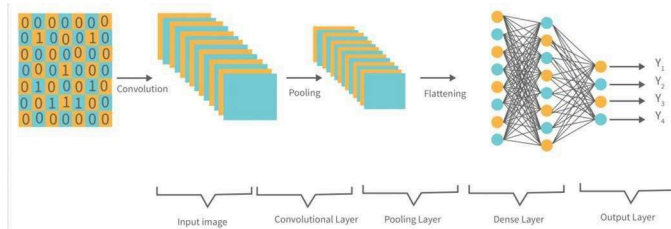


Fig. 3. Architecture of CNN

begun with certain preliminary hyper parameters. CNNs have recently become more popular, and DL is the most widely used architecture since DL models can learn important features from input images at various convolutional levels like a human brain does. With excellent classification accuracy and a low error rate, DL can resolve complex problems extremely effectively and efficiently [11]. The most prominent parts of a CNN include convolutional layers, pooling layers, and fully connected layers. Convolutional layers use a filter set to scan the input data, with each filter set designed to pick up on a specific pattern or feature of the data. The most prominent parts of a CNN include convolutional layers, pooling layers, and fully connected layers. The pooling layers decrease the dimensionality of the data by downsampling the output of the convolutional layers. The fully connected layers merge the features identified by the convolutional and pooling layers and utilize them to make predictions regarding the input data. The major benefit of CNNs is that they can learn to automatically identify and extract features from the input data,

Epoch 1/10	517/517	1968s 4s/step	- accuracy: 0.5185 - loss: 1.5164 - val_accuracy: 0.8231 - val_loss: 0.5496
Epoch 2/10	517/517	1935s 4s/step	- accuracy: 0.8528 - loss: 0.4346 - val_accuracy: 0.8149 - val_loss: 0.5542
Epoch 3/10	517/517	1935s 4s/step	- accuracy: 0.9197 - loss: 0.2355 - val_accuracy: 0.8811 - val_loss: 0.3716
Epoch 4/10	517/517	1970s 4s/step	- accuracy: 0.9597 - loss: 0.1248 - val_accuracy: 0.8879 - val_loss: 0.3860
Epoch 5/10	517/517	1963s 4s/step	- accuracy: 0.9793 - loss: 0.0664 - val_accuracy: 0.8802 - val_loss: 0.4433
Epoch 6/10	517/517	1951s 4s/step	- accuracy: 0.9778 - loss: 0.0657 - val_accuracy: 0.8828 - val_loss: 0.4842
Epoch 7/10	517/517	1940s 4s/step	- accuracy: 0.9756 - loss: 0.0762 - val_accuracy: 0.8862 - val_loss: 0.4614
Epoch 8/10	517/517	1943s 4s/step	- accuracy: 0.9894 - loss: 0.0303 - val_accuracy: 0.8661 - val_loss: 0.5476
Epoch 9/10	517/517	1893s 4s/step	- accuracy: 0.9884 - loss: 0.0393 - val_accuracy: 0.8557 - val_loss: 0.6371
Epoch 10/10	517/517	1873s 4s/step	- accuracy: 0.9839 - loss: 0.0544 - val_accuracy: 0.8826 - val_loss: 0.5558
	129/129	133s 1s/step	

Fig. 4. Summary of the trained CNN Model

It lessens the requirement for feature engineering by hand. This makes them ideal for tasks like segmentation, object detection, and image classification.

The architecture of the CNN model constructed for image classification is illustrated in Fig 4. CNN is a form of deep learning neural network used to handle data that is structured with a grid-like arrangement.

It is composed of several layers of connected nodes, with each one taking a particular mathematical operation on the input data. The Architecture of the CNN is depicted in Fig 3. It is to be noted that training a neural network for a limited number of epochs is but a part of the entire training process. To get good performance, it is also important to select and preprocess the training data properly, and to adjust the hyperparameters of the network in order to maximize its performance. Further, the performance of the network could be enhanced if it were trained for a larger number of epochs, but this would also increase the amount of time and computational power needed for training.

IV. RESULTS

Leaf disease identification is the process of classifying the disease affecting a plant's leaves. It is normally accomplished by examining the leaves visually and noting the symptoms they show, like discoloration, spotting, or wilting. The symptoms observed are then used to identify the disease that exists. The output of using the Canny edge detection algorithm on an image is a binary image where pixels that map to edges in the original image are labeled as "on" (typically stored as white pixels), and all the other pixels are labeled as "off" (typically stored as black pixels). The resulting image can subsequently be used for subsequent image processing tasks, like object recognition or image segmentation. The output of canny edge algorithm is depicted in Fig 5



Fig. 5. (a) Input image (b) result of image thresholding

The outcome of using image thresholding is a binary image that retains the significant characteristics of the original image but removes most of the noise and decreases the amount of data that has to be processed. The outcome of image thresholding algorithm is demonstrated in Fig 5

This can be applied to a wide range of image processing operations, including object recognition or image segmentation. By the 10th epoch of training, a CNN will have learned to identify and classify objects in images with a high level of accuracy. The actual performance of the network would vary depending upon various factors like the quality of training data as well as their diversity, specific network architecture used, and used hyperparameters to train the model. Generally speaking, however, a properly trained CNN should learn to have excellent accuracy on very diverse image classification tasks after only 10 rounds of training. Training results appear in Fig 7 It's necessary

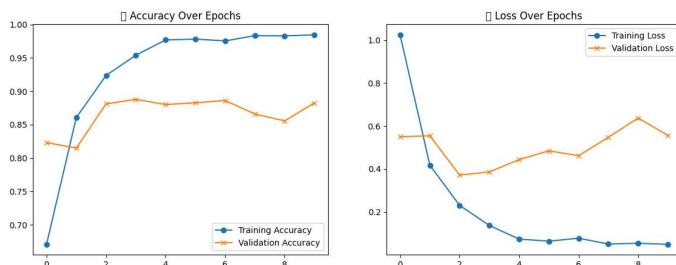


Fig. 6. Graph depicting training and validation accuracy and loss

to point out that training a neural network to a specific number of epochs is only a part of the process of training. To get a good performance, it's also important to choose and preprocess the training data well, and adjust the network's hyperparameters to get the best performance. In addition, the performance of the network can be enhanced if the network is trained for more epochs, but this will also boost the training time and computational resources needed. A confusion matrix is a table used to measure the performance of a classification model, most often in the context of supervised machine learning. The matrix is utilized to measure the model's capacity to accurately classify data into one of multiple classes, and is frequently utilized in combination with classification metrics like accuracy,

precision, and recall. Confusion matrix of above training is presented in Fig 8 A confusion matrix for a 15-class classification model would have 15 rows and 15 columns where each row would correspond to the predicted class and each column would correspond to the actual class. The matrix entries would indicate how many instances within the test data were predicted to belong to some class and truly do belong to that class and how many predicted instances belonged to a certain class but actually are members of another class.

V. CONCLUSION

We infer from this work that disease identification and classification are possible through image processing. The proposed method can classify crop diseases very accurately and efficiently. The proposed method was created with farmers and the agricultural industry in consideration. The technology developed will identify plant disease and suggest corrective measure. Proper information of the disease and the therapy can be utilized to improve the health of the plant. It is possible to use digital image pro-

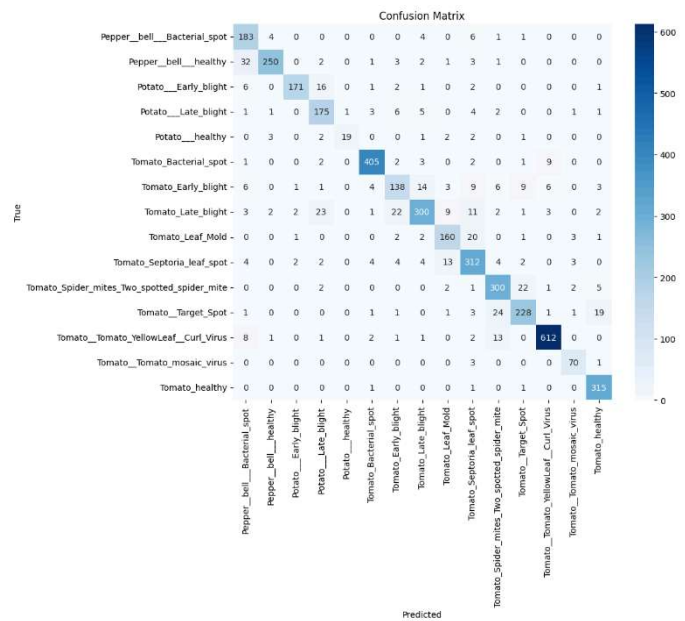


Fig. 7. (Confusion matrix of predicted classes

Processing methods to carry out image classification of sick leaves of 15 classes. Through training a proper machine learning model on a labeled dataset of leaf images, one can build a system capable of automatically recognizing and classifying the various kinds of diseases that occur in the leaves. The accuracy of the classification model will be influenced by a variety of factors, including the diversity and quality of the training data, the machine learning algorithm employed, and the hyperparameters of the model. Broadly speaking, however, an appropriately trained model should be capable of generating high accuracy on a range of image classification tasks. Applying digital image processing methods to classify infected leaves has a variety of advantages that range from enhanced diagnostic precision, speed, and efficiency in detecting diseases, to automating the classification process. This saves time and resources when detecting and treating plant diseases, and ultimately leads to better health and productivity in agricultural crops.

REFERENCES

1. YASHWANT KURMI, PRANKUR SAXENA, BHUPENDRA KIRAR, SUCHI GANGWAR, VIJAYSHRI CHAURASIA, AND ADITYA GOEL. DEEP CNN MODEL FOR CROPS' DISEASES DETECTION USING LEAF IMAGES. *MULTIDIMENSIONAL SYSTEMS AND SIGNAL PROCESSING*, 33, 09 2022.
2. MR KALE. Analysis of crop disease detection with svm, knn and random forest classification. *INFORMATION TECHNOLOGY IN INDUSTRY*, 9:364–372, 03 2021.
3. Colin Thirtle, Xavier Irz, Lin Lin, Victoria McKenzie-Hill, and Steve Wiggins. Relationship between changes in agricultural productivity and the incidence of poverty in developing countries. *report commissioned by the Department for International Development, London*, 2001.
4. Priya Ujave and Smita Nirkhi. *Comparative Study of Tomato Crop Disease Detection System Using Deep Learning Techniques*, pages 493–
5. Radhika Bhagwat and Yogesh Dandawate. A framework for crop disease detection using feature fusion method. *International Journal of Engineering and Technology Innovation*, 11:216–228, 06 2021.
6. V. Devi, R. Prabavathi, P. Subha, and M. Meenaloshini. An efficient and robust random forest algorithm for crop disease detection. pages 1–4, 03 2022.
7. Darrel S Metcalfe, Donald M Elkins, and Harold De Mott Hughes. *Crop production: principles and practices*. Macmillan, 1980.
8. Fraol Gelana, Taye Debelee, Friedhelm Schwenker, Yehualashet Ayano, and Samuel Rahimeto. Machine learning in cereal crops disease detection: A review. *Algorithms*, 15:75, 02 2022.
9. Jai Vardhan IIIT Naya Raipur *Detection of Healthy and diseased crops in drone captured images using deep learning 2023*.
10. Bhakti Thakre. Clever farming with crops disease detection system using iot and ml. *International Journal of Engineering Applied Sciences and Technology*, 7:302–307, 05 2022.
11. Mohammadi Iram. Crop disease detection by machine learning. *International Journal for Research in Applied Science and Engineering Technology*, 8:1096–1101, 08 2020.